Software Design Diagram Generator

Group No. 27

Final Stakeholder and Elicitation Methods

1. Stakeholders

S1 — Software Developer / Engineer

Stakeholder type: Primary end user

Role in the system: Uses requirements to generate and inspect design diagrams.

Elicitation methods: EM1 Questionnaire (Google Form), EM4 Semi-structured interview (conditional)

What was gathered:

• Pain points: diagramming is time-consuming, hard to translate requirements into elements, hard to keep consistent with changing requirements.

• Preferred inputs: typed/pasted text, PDF/DOCX SRS, user stories, FR/NFR lists and multiple documents.

• Generated diagrams are treated as drafts or starting points that must be reviewed and edited.

• Expected editing: add/remove, move, rename, edit relationships, regenerate selected parts or the whole diagram.

Status: Survey with 31 responses (14 Oct 2026). Interviews not conducted.

S2 — Software Architect / Designer

Stakeholder type: Primary end user

Role in the system: Uses generated diagrams for architectural and design decisions.

Elicitation methods: EM1 Questionnaire (Google Form); EM2 Document analysis of UML/DFD notation standards

What was gathered:

• Diagram types needed: Use Case, Class, Activity, Sequence, DFD and Component.

• Initial priority from the survey: Use Case, Activity and Sequence.

• Notation must follow OMG UML and Gane-Sarson/Yourdon DFD conventions.

• Unsupported elements must be marked as assumptions, and every element must trace back to a requirement.

S3 — Project Manager / Team Lead

Stakeholder type: Secondary user

Role in the system: Uses diagrams and requirement information for planning, review and communication.

Elicitation methods: EM3 Internal team brainstorming (facilitated by the Product Owner / Team Lead)

What was gathered:

• Need a project-level coverage summary and a shareable combined report.

• Need an activity log showing who changed requirements and diagrams, and when.

• Need to invite collaborators and assign roles.

S4 — System Administrator

Stakeholder type: Administrative stakeholder

Role in the system: Manages users, system configuration and access where applicable.

Elicitation methods: EM3 Internal team brainstorming; EM2 Document analysis of AI provider and hosting documentation

What was gathered:

• Account management: view, add, deactivate and reactivate accounts.

• Monitoring: active users, projects, storage against limits, error logs, generation-time statistics.

• Free-tier AI provider and hosting limits that the system must handle (rate limits, backend sleep, storage cap).

S5 — Development Team

Stakeholder type: Development stakeholder

Role in the system: Builds, tests, maintains and evolves the system.

Elicitation methods: EM3 Internal team brainstorming; EM2 Document analysis of course instructions and the Lab 6 brief

What was gathered:

• Project-specific interpretation of the brief, feasibility and prioritisation into four sprints.

• Technology constraints: Gemini free tier, Vercel frontend, Render backend, free-tier storage and email.

• Scope decisions such as English-only input, no OCR, and Activity diagram in place of a separate flowchart.

2. Elicitation Methods

EM1 — Questionnaire / Google Form

Applied to: S1 Software Developer / Engineer, S2 Software Architect / Designer (students and developers)

Format: Online Google Form

What was gathered:

• 31 responses (all undergraduate students).

• Ambiguity handling: ask for clarification, highlight the requirement and explain why it was flagged.

• Change propagation: most respondents want related diagrams updated after confirmation.

• After generation: explanation of derivation, traceability, list of assumptions, validation warnings.

• Acceptable generation time: mostly 10–60 seconds.

• Security: login, encrypted transfer/storage, limited retention of documents.

• Export: PNG, SVG, PDF, editable XML and PlantUML.

EM2 — Document Analysis

Applied to: Course instructions, UML/DFD notation standards (S2), AI provider and hosting documentation (S4)

Format: Async, individual review

Owning sub-team: Requirements/Docs (course material), DFD Renderer (notation), AI/Extraction (AI provider), App Shell/Infra (hosting)

What was gathered:

• Required deliverables, terminology and course constraints.

• Notation rules for each diagram type (DR1).

• AI provider rate limits and terms, hosting sleep behaviour and non-persistent disk (C1–C3).

EM3 — Internal Team Discussion / Brainstorming

Applied to: S3 Project Manager / Team Lead, S4 System Administrator, S5 Development Team

Format: Whole team

What was gathered:

• Interpretation of the brief, feasibility and prioritisation.

• Roles and permissions (Viewer, Editor, Owner, System Administrator).

• Resolution of conflicts between requirement drafts.

Status: Ongoing;